

Practical 2 – The Sensor Voltage Divider

Booklet 3, section 10, page 23. Success criteria 4.3, 4.4 and 4.5.

Everything you write goes in your jotter. Copy each numbered table before you start that section, and answer each question by its number.

The idea being tested

Swap one resistor in a voltage divider for a sensor and V_2 follows whatever the sensor senses. Where the sensor sits – at the top or at the bottom – decides whether V_2 rises or falls.

Light dependent resistor (LDR): bright light → low resistance. Dark → high resistance.

NTC thermistor: hot → low resistance. Cold → high resistance.

V_2 is the voltage that switches the transistor in Practical 3, so it is the only voltage you measure here. Exam papers call it V_{in} or V_{out} .

2.1 What you are testing

Aim: to find out how V_2 changes with light and with temperature, and how moving the sensor from the top to the bottom changes that.

Four circuits, each on a 5 V supply with a 10 k Ω fixed resistor as the sensor's partner: the LDR at the top, the LDR at the bottom, the thermistor at the top and the thermistor at the bottom.

2.2 Predict

Table 2.2 – which way will V_2 go?

Circuit	Change	Sensor resistance (rises / falls)	V_2 (rises / falls)
LDR at the top	getting darker		
LDR at the bottom	getting darker		
thermistor at the top	getting hotter		
thermistor at the bottom	getting hotter		

Q1 A street light has to come on when it gets dark, so it needs a V_2 that rises as the light fades. Use your table to state which LDR position does that.

[1]

2.3 Simulate in Yenka

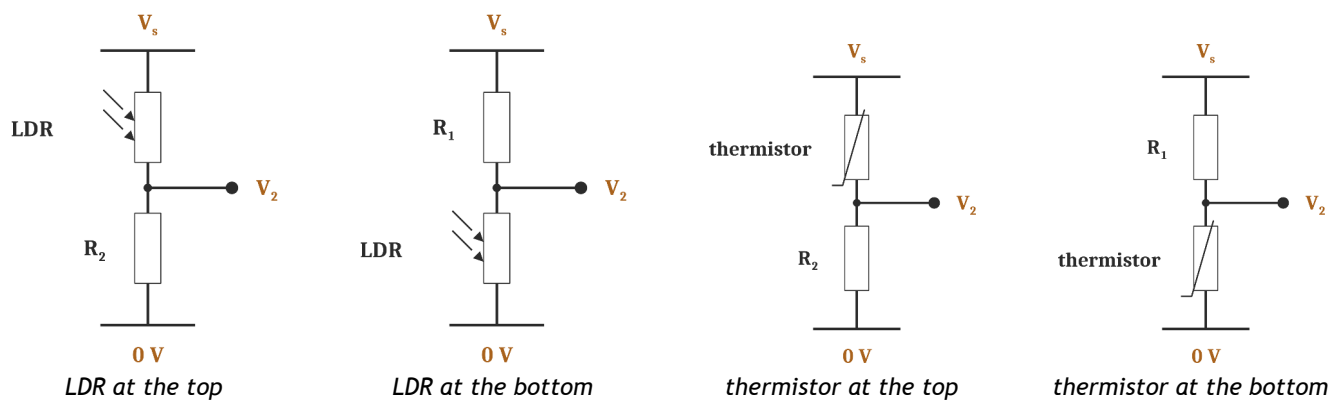


Table 2.3a – simulated V_2 , LDR at the top and bottom

Light level	V_2 , top (V)	V_2 , bottom (V)
very bright		
room light		
dim		
very dark		

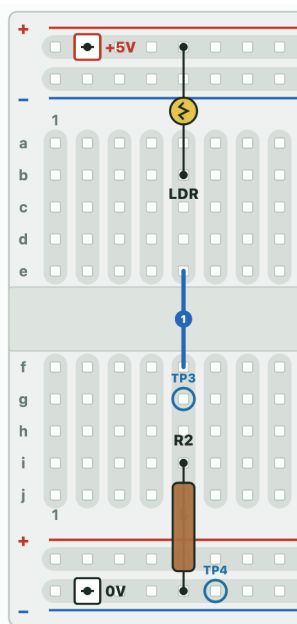
Table 2.3b – simulated V_2 , thermistor at the top and bottom

Temperature	V_2 , top (V)	V_2 , bottom (V)
cold		
room temperature		
warm		
hot		

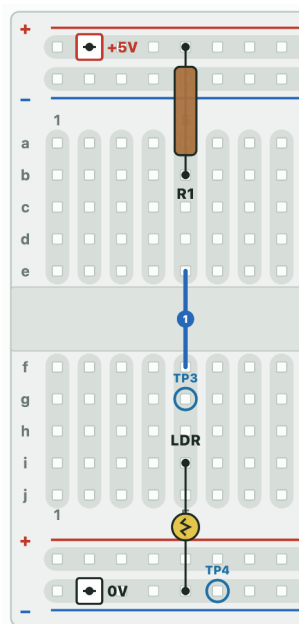
Q2 Compare the two columns in each table. Describe what moving the sensor from the top to the bottom does to the way V_2 changes. [2]

2.4 Build on the breadboard

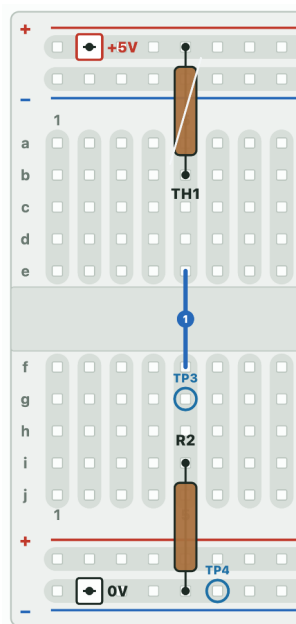
On every board the sensor, the jumper and the 10 k Ω resistor run down column 5. Measure V_2 between TP3 and TP4.



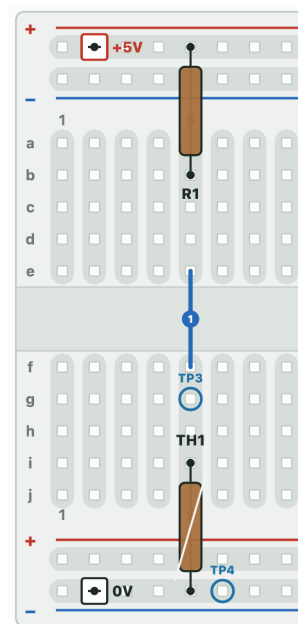
LDR at the top



LDR at the bottom



thermistor at the top



thermistor at the bottom

Table 2.4a – measured V_2 , LDR at the top and bottom

Light level	V_2 , top (V)	V_2 , bottom (V)
bright (bench lamp close)		
room light		
shaded by hand		
covered (dark)		

Table 2.4b – measured V_2 , thermistor at the top and bottom

Thermistor	V_2 , top (V)	V_2 , bottom (V)
cooled with ice in a sealed bag		
room temperature		
held between finger and thumb		

2.5 Compare

Write rises or falls in each box.

Table 2.5 – predicted, simulated and measured

Circuit and change	Predicted V_2	Simulated V_2	Measured V_2
LDR at the top, getting darker			
LDR at the bottom, getting darker			
thermistor at the top, getting hotter			
thermistor at the bottom, getting hotter			

Q3 Did the real circuits do what you predicted in every row? If not, suggest a reason. [2]

Q4 Describe how the circuit with the LDR at the bottom works as it gets darker. Refer to the resistance of the LDR and V_2 . [2]

Q5 Use your room-light reading for the LDR at the top. Calculate the voltage across the LDR, then the resistance of the LDR. [4]

2.6 What the results tell you

Q6 Explain why the partner resistor is 10 k Ω and not 100 Ω . [2]

Q7 The 10 k Ω resistor is replaced with a variable resistor. Explain the effect this has on the circuit. [2]

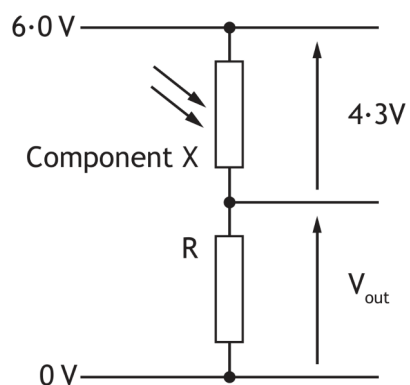
Q8 Draw a sensing circuit whose V_2 increases when it is cold. [2]

Table 2.6 – test plan for your circuit

Test	Planned test	Expected result (in terms of V_2)
Test 1		
Test 2		

Q9 Copy Table 2.6 and complete it for your circuit: one test that makes it colder and one that makes it warmer. [3]

From a past exam paper



Light levels at a weather monitoring station are measured using this sensing sub-system.

Sources & credits: © Qualifications Scotland (SQA). National 5 Engineering Science Specimen Question Paper, Question 15.

Q10 State the full name of Component X. [1]

Q11 Calculate the value of V_{out} . [1]

Q12 Explain the effect on V_{out} of increasing and decreasing light level. [3]